

Physiology-Constrained Wavelet Kolmogorov–Arnold Networks for Inter-Patient ECG Arrhythmia Classification

Venkateramanan Manivannan and Ramanathan Lakshmanan

Abstract—Beat-level arrhythmia classifiers are frequently evaluated under protocols that place beats from the same patient in both training and test sets, and their interpretability is typically asserted from filter visualisations rather than measured. We present PC-WavKAN, a 153,045-parameter wavelet Kolmogorov–Arnold Network for inter-patient ECG classification. It extends Wav-KAN with physiology-constrained wavelet initialisation (PCWI), which initialises each output-channel group’s translation and dilation from QRS, P-wave and T-wave morphological priors, and trains with a Rarity-Anchored Curriculum (RAC) annealing from class-balanced to natural sampling. Evaluation follows the de Chazal DS1/DS2 inter-patient protocol on MIT-BIH using 20 seeded runs per arm, paired Wilcoxon signed-rank tests, Holm–Bonferroni correction and Cohen’s d . The architecture was selected on DS1 validation alone, where the adopted configuration outperformed all six alternatives tested (Holm-adjusted $p \leq 0.0008$); the component ablation identifies PCWI as the only novel component with a corrected large effect there, and shows the mother wavelet to be interchangeable. On held-out DS2, Macro-F1 is 0.357 ± 0.019 : statistically indistinguishable from four recipe-matched baselines of 29K–118K parameters (all Holm-adjusted $p \geq 0.36$, every paired 95% confidence interval including zero). RAC yields a large gain in supraventricular recall ($0.137 \rightarrow 0.222$, Holm-adjusted $p < 10^{-4}$, $d = +1.47$) and in fusion recall. Benchmarked against an untrained null, PCWI’s physiological structure is substantially retained after training; we report this as prior retention, not learned alignment. Cross-dataset, PC-WavKAN attains the highest INCART Macro-F1 of five models but is weaker on SVDB, which per-class analysis attributes to a supraventricular-to-ventricular bias shared with B-spline KANs.

Index Terms—Kolmogorov–Arnold Networks, Wavelets, Arrhythmia Classification, Inter-Patient Evaluation, Curriculum Learning, Class Imbalance, Model Transparency

Manuscript submitted for review to the IEEE Journal of Biomedical and Health Informatics.

Venkateramanan Manivannan is with the School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology (VIT), Vellore, Tamil Nadu 632014, India (e-mail: venkate.ramanan2024@vitstudent.ac.in).

Ramanathan Lakshmanan is a Professor with the Department of IoT, School of Computer Science and Engineering, Vellore Institute of Technology (VIT), Vellore, Tamil Nadu 632014, India (Corresponding author, e-mail: lramanathan@vit.ac.in).

I. INTRODUCTION

CARDIOVASCULAR disease is the leading cause of death worldwide, and continuous cardiac rhythm monitoring is central to its management. The electrocardiogram (ECG) remains the reference modality for arrhythmia detection, but manual interpretation of long-term Holter recordings is labour-intensive and does not scale to population screening. Convolutional and attention-based deep networks now report high beat-classification accuracy [1]–[3].

Two properties of that literature limit what those numbers establish. First, much of it evaluates under an *intra-patient* protocol, in which beats from a single recording are distributed randomly across training and test partitions. Because a classifier can then memorise a patient’s baseline morphology, accuracies above 97% are routinely reported that do not transfer to unseen patients [4]–[6]. Under the strict inter-patient protocol of de Chazal *et al.* [7], performance drops sharply, and minority-class recall in particular collapses. Second, transparency claims in this literature are usually qualitative. Models are described as interpretable on the basis of a figure of learned filters, or by attaching a post-hoc attribution method such as Grad-CAM whose output is itself sensitive to noise and to the choice of attribution algorithm [8].

Kolmogorov–Arnold Networks (KANs) [9] respond to the second problem: their learnable univariate functions sit on network *edges* rather than at nodes, so the learned function space is inspectable in parameter space rather than only through gradients or occlusion. Bozorgasl and Chen’s Wav-KAN [10] replaces the original B-spline basis with wavelets parameterised by translation and dilation, aligning the basis with classical time–frequency analysis of the ECG [11]. Applications of KANs to ECG such as MAK-Net [12] show competitive accuracy, but the inspectability of the basis has not been converted into a measured, falsifiable quantity: an inspectable parameter is not the same thing as one that demonstrably encodes physiology.

This work addresses a specific gap at the intersection of those two problems. The wavelet-KAN basis is parameterised in exactly the units — a temporal/amplitude offset and a width — in which ECG fiducial morphology is conventionally de-

scribed. That correspondence can be exploited at initialisation and then *measured* after training. Related work is reviewed in §II. Our contributions are:

- 1) **PC-WavKAN, a physiology-constrained wavelet KAN.** We introduce physiology-constrained wavelet initialisation (PCWI), which partitions the KAN’s output channels into QRS-, P- and T-associated groups and initialises each group’s wavelet translation and dilation from morphological priors for that component, replacing Wav-KAN’s isotropic random initialisation. A component ablation resolved entirely on DS1 validation data identifies PCWI as the only novel component of the architecture with a large, multiplicity-corrected positive effect, and shows that the choice of mother wavelet within the family is not measurably consequential (§V-A). That ablation was run against the base configuration from which the final architecture was selected, so PCWI’s contribution is established relative to that base rather than re-measured on the adopted model; we state this scope wherever the claim is used and treat closing it as the study’s main outstanding experiment (§VI-E).
- 2) **Measured retention of the physiological prior, with an explicit null.** We define Physiological Prior Retention (PPR) as the normalised proximity of the trained wavelet parameters to those priors, and — unlike interpretability scores reported without a reference point — we evaluate it against both an untrained PCWI model and a randomly initialised one. This shows what the measurement does and does not license: the trained basis retains physiological structure far beyond a random basis, but the score decreases from its initialisation value, so it quantifies *retention*, not learned alignment (§V-E).
- 3) **A curriculum that targets the class the inter-patient protocol breaks.** The Rarity-Anchored Curriculum (RAC) anneals from class-balanced to natural sampling. On the adopted architecture it produces a large, corrected improvement in supraventricular (S) recall and fusion (F) recall at no significant cost to other classes (§V-C).
- 4) **An evaluation protocol strong enough to support a negative result.** All comparisons use 20 independently seeded runs, paired tests, Holm–Bonferroni correction within declared families, effect sizes, and confidence intervals; the architecture was selected without reference to the test set. Under that protocol the honest finding on the field’s headline metric is parity, not superiority, and we report it as such (§V-B).

The resulting position is deliberate: a model that matches four standard architectures on Macro-F1 while additionally supporting a measurable structural property and a targeted minority-class gain is a different contribution from an accuracy win, and the diagnostic findings this protocol produced are results in their own right.

II. RELATED WORK

A. Inter-Patient Evaluation

Systematic reviews by Xiao *et al.* [5] and Silva *et al.* [6] document how prevalent intra-patient evaluation remains and how strongly it inflates reported performance; Bahrami *et al.* [4] quantify the resulting optimism directly. The DS1/DS2 partition of de Chazal *et al.* [7] is the established remedy for MIT-BIH and is the protocol we adopt without modification. Architectures evaluated under it include the CNN–RNN hybrid of Guo *et al.* [2], the multiscale attention CNN of Zhou *et al.* [13], and the sequence-to-sequence models of Mousavi *et al.* [14]. We do not tabulate published accuracies from these works against ours — differences in lead selection, beat extraction, AAMI mapping and Q-class handling make them non-comparable — and instead retrain four architectures ourselves under one protocol (§IV-E).

B. Kolmogorov–Arnold Networks and Wavelet Bases

KANs [9] place learnable univariate functions on edges, with B-splines as the original basis. Wav-KAN [10] substitutes wavelet functions — including the Mexican Hat — parameterised by translation and dilation, arguing that this improves both multi-resolution capacity and interpretability. Our architecture inherits this construction: the wavelet-KAN layer, the Mexican Hat basis, and the argument that a wavelet basis is more inspectable than an opaque activation are due to [9], [10] and are not claimed here. MAK-Net [12] pairs spline-activation KAN layers with multiscale convolution and a BiGRU for imbalanced ECG classification, reporting interpretability via Grad-CAM — post-hoc attribution over inputs rather than a measurement on the basis itself.

What we add to this line of work is not a new basis but a constraint on it and a measurement of that constraint’s persistence. Initialisation of KANs is itself an active question: Rigas *et al.* [15] study LeCun- and Glorot-inspired and power-law schemes for KAN basis coefficients, and find the choice materially affects stability and accuracy. That work is deliberately basis- and domain-agnostic and optimises a numerical property — activation-variance preservation across layers. PCWI is orthogonal to it and complementary: it leaves the coefficient initialisation to such a scheme and instead sets the *translation and dilation* of each channel group from domain priors, which is possible only because those two parameters have units a domain expert can specify. To our reading of this literature, wavelet-KAN translation and dilation have not previously been initialised from ECG fiducial priors, nor has the post-training distance between learned basis parameters and such priors been reported for any KAN-based ECG classifier. We deliberately state this as a gap in reported practice rather than a claim of priority, and we make the accompanying measurement falsifiable by publishing its null baselines (§V-E).

C. Class Imbalance and Curricula

Inter-patient MIT-BIH exhibits N:F ratios above 100:1 [6], under which plain cross-entropy training tends toward majority-class collapse. Curriculum learning [16] varies the

sampling distribution or loss weighting over training. Bae and Shin [17] show naive contrastive training on imbalanced medical data can *degrade* performance while a curriculum-phased variant recovers it, and Schmale *et al.* [18] that prioritising rare phenotypes stabilises decision boundaries. Our curriculum is of this family, and §fsec:results reports where its benefit is and is not statistically robust. A methodologically distinct alternative operates on logits rather than on sampling: Dai *et al.* [19] apply adaptive logit adjustment to long-tailed *record-level* multi-label ECG diagnosis, learning bounded label-state-specific prior corrections. That is a different task granularity from the beat-level single-label problem addressed here, so we do not treat it as a comparator, but it is a concrete alternative worth testing at beat level (§VI-D).

D. Efficient Inference

Farag [20] and Elsheikhy *et al.* [21] argue for small memory footprints for on-device ECG inference. We report measured parameter counts, model size, x86 CPU latency and post-training quantisation behaviour for our own model (§V-I) rather than inferring efficiency from parameter count alone, and we are explicit about the hardware class on which those measurements were taken.

III. METHOD

A. Problem Formulation

Each annotated beat is represented by a signal window $x_i \in \mathbb{R}^T$ ($T = 360$ samples, 360 Hz, centred on the annotated R-peak) and a five-element inter-beat-interval vector $r_i \in \mathbb{R}^5$. The task is to learn $f : (x_i, r_i) \mapsto y_i$ with $y_i \in \{N, S, V, F, Q\}$ under severe imbalance ($N \gg V > S > F \gg Q$).

Temporal scope of the task: Two aspects of this representation make the task *offline beat classification* rather than streaming classification, and we state them here because they bound every claim in this paper. First, r_i contains two intervals that follow beat i (§III-B.4), so labelling beat i requires observing beats $i+1$ and $i+2$. Second, the signal is band-pass filtered over each complete record before beat extraction, which is not a causal operation. A deployment of this model would therefore require a two-beat lookahead buffer and a causal replacement for the record-level filter. §V-D shows the lookahead is not a formality: the model draws statistically significant signal from one of the two following intervals. We accordingly avoid the terms real-time and streaming throughout.

B. PC-WavKAN Architecture

The model (Fig. 1) has a morphology branch and a rhythm branch, fused before classification.

1) Wavelet-KAN Morphology Backbone: Following WavKAN [10], the learnable function sits on every edge (j, k) rather than at the node (Fig. 2): each edge carries an independently parameterised Mexican Hat wavelet ψ , the negative normalised second derivative of a Gaussian. The layer maps $x_i \in \mathbb{R}^{360}$ to $z_{\text{kan}} \in \mathbb{R}^{64}$,

$$z_{\text{kan}}^{(k)} = \sum_{j=1}^{360} w_{j,k} \psi\left(\frac{x_j - \mu_{j,k}}{\gamma_{j,k}}\right) + \lambda \sum_{j=1}^{360} v_{j,k} x_j, \quad (1)$$

for $k = 1, \dots, 64$, where $\mu_{j,k}$ and $\gamma_{j,k}$ are the learnable translation and dilation of edge (j, k) , $w_{j,k}$ its weight, and the second term is a linear residual path with a fixed gate $\lambda = 0.1$ that we retain for gradient flow. The output is layer-normalised and passed through a bidirectional GRU (32 hidden units per direction) yielding $z_m \in \mathbb{R}^{64}$.

2) Physiology-Constrained Wavelet Initialisation (PCWI): PCWI (§III-B.2) is the component that distinguishes this architecture from a generic wavelet KAN. The 64 output channels are partitioned into three groups associated with the ECG components whose morphology the branch must separate, and each group's edge parameters are initialised from a prior for that component rather than from a single isotropic distribution:

$$\mu_{j,k} \sim \mathcal{U}(\bar{\mu}_c - \epsilon_c, \bar{\mu}_c + \epsilon_c), \quad \gamma_{j,k} \sim \mathcal{U}(\bar{\gamma}_c - \delta_c, \bar{\gamma}_c + \delta_c) \quad (2)$$

for every edge of the channel group c , with $(\bar{\mu}_c, \bar{\gamma}_c)$ set from morphology and (ϵ_c, δ_c) small jitter terms that break symmetry between edges. PCWI constrains only the translation and dilation; the edge coefficients $w_{j,k}$ keep the standard variance-preserving initialisation [15], so PCWI is a domain-prior constraint layered on top of a numerical scheme rather than a replacement for one. Parameters are in units of the per-beat z -scored signal (§IV-A), in which the input is approximately zero-mean and unit-variance. We use channels 1–32 for QRS with $(\bar{\mu}, \bar{\gamma}) = (0.0, 0.05)$, since the QRS complex crosses the beat's own baseline and is the sharpest transient, so a narrow wavelet is appropriate; channels 33–48 for the P-wave with $(-0.10, 0.12)$; and channels 49–64 for the T-wave with $(+0.10, 0.10)$, both being slower, lower-amplitude deflections and therefore wider.

We are deliberate about what (2) is. It is a physiologically *motivated* allocation of the initial function space, not an assertion that channel group c computes component c : the priors constrain where in parameter space each group begins, they are not constraints during training, and μ and γ are amplitude-domain quantities that correspond to ECG components by way of those components' characteristic amplitude and sharpness rather than by explicit temporal localisation. §V-A tests whether the allocation helps and §V-E tests whether it survives training.

3) Gated P-Region Fusion: Supraventricular beats are distinguished largely by P-wave abnormality (shortened PR interval, inverted or absent P-wave), but the P-wave contributes a small fraction of a beat window's total energy relative to the QRS and T-wave. We therefore add a low-parameter pathway over the pre-R interval: samples 80–160 of the window, i.e. 278 ms to 56 ms before the R-peak, are encoded by a small randomly initialised wavelet-KAN layer, projected to the morphology dimension, and combined with z_m through a learned sigmoid gate that determines per beat how much P-region information enters the main representation. The pathway is implemented with a multi-head attention block, but as it attends over a single key/value token, the softmax is degenerate and the operation reduces to a learned linear projection; the selectivity is supplied by the gate, not by attention weights, and we describe it as gated fusion rather than as attention for that reason. It adds approximately 34K parameters.

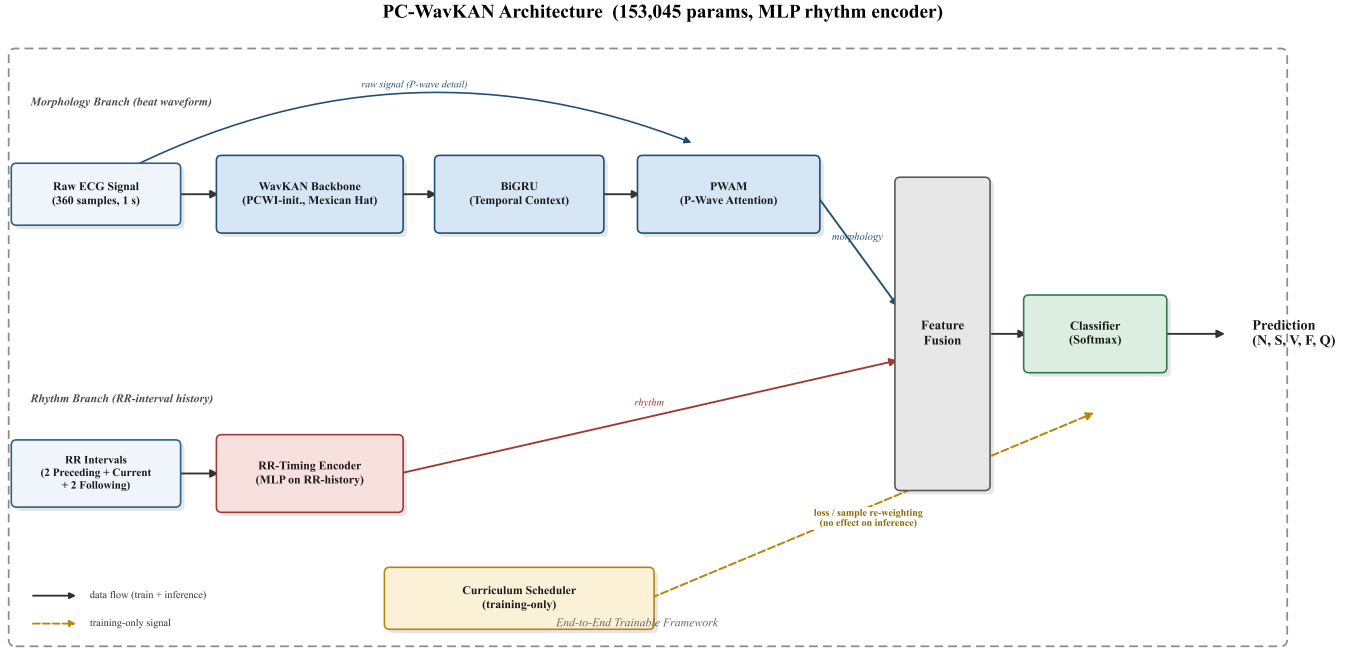


Fig. 1. **PC-WavKAN**. The morphology branch (PCWI wavelet-KAN layer → BiGRU temporal context → gated P-region fusion, which takes the raw beat as a second input alongside the BiGRU output) is concatenated with the rhythm branch before classification. The curriculum scheduler acts on training only.

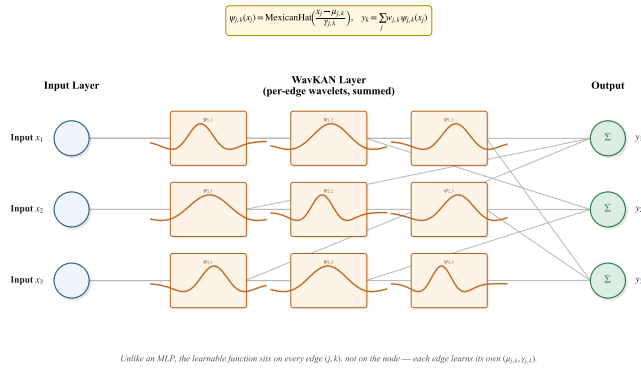


Fig. 2. **Wavelet-KAN edge parameterisation**. Learnable wavelets $\psi(t; \mu, \gamma)$ replace fixed node activations; every edge of the $j \times k$ grid carries its own translation and dilation, as in (1).

4) Rhythm Branch: The rhythm vector is

$$r_i = [RR_{-2}, RR_{-1}, RR_0, RR_{+1}, RR_{+2}], \quad (3)$$

where RR_0 is the interval from beat $i-1$ to beat i (the pre-beat interval of the classified beat), RR_{-1} and RR_{-2} are the two intervals preceding it, and RR_{+1} and RR_{+2} are the two intervals following beat i . Intervals are in seconds and clipped to $[0.2, 3.0]$ s to bound the effect of missed or spurious detections; no further normalisation is applied. The vector is encoded to $z_r \in \mathbb{R}^{16}$. Two encoders were evaluated: a single-head self-attention block over the five intervals projected to tokens, and a plain multi-layer perceptron over the raw vector. §V-A selects between them.

5) **Fusion and Classification**: The concatenation $z = [z_m; z_r] \in \mathbb{R}^{80}$ is classified by an $80 \rightarrow 48 \rightarrow 5$ head. With the

self-attention rhythm encoder the model has 154,325 trainable parameters; with the plain-MLP encoder selected in §V-A, 153,045. Both are more than an order of magnitude smaller than MAK-Net (6.1M) [12] and comparable dense recurrent architectures [2]. All results outside §V-A use the 153,045-parameter configuration.

C. Rarity-Anchored Curriculum (RAC)

RAC restructures training by class rarity in two phases (Algorithm 1). During warm-up (the first 25% of the epoch budget) mini-batches are drawn by a fully class-balanced sampler and the loss is unweighted cross-entropy, so that no class's decision boundary is established purely by frequency before the rare classes have been seen. During annealing the sampler's balancing exponent β decays linearly from 1 (balanced) to 0 (natural), paired with focal loss ($\gamma = 2$) under inverse-frequency class weights, with the S-class weight multiplied by 8 to counter its morphological similarity to N. The schedule affects training only; inference always uses the natural distribution. We name it *rarity-anchored* rather than *progressive* to distinguish it from difficulty-ordered curricula: the ordering variable is class frequency, which is known in advance, not per-example difficulty, which is not.

D. Physiological Prior Retention (PPR)

To make the effect of PCWI measurable rather than illustrative, we quantify how close the trained wavelet parameters remain to the priors of (2). For channel group c with prior

Algorithm 1 Rarity-Anchored Curriculum (RAC)

```

1: Input: training set  $\mathcal{D}$ , epoch budget  $E$ , warm-up fraction  $\alpha = 0.25$ 
2: Output: trained model  $f_\theta$ 
3: Compute class counts  $\{n_c\}$ ,  $c \in \{N, S, V, F, Q\}$ ; inverse-frequency weights  $w_c$ ;  $w_S \leftarrow 8w_S$ 
4: for epoch  $e = 1$  to  $E$  do
5:   if  $e \leq \alpha E$  then
6:      $\beta \leftarrow 1$ ;  $\mathcal{L} \leftarrow \mathcal{L}_{CE}$  (unweighted)
7:   else
8:      $\beta \leftarrow 1 - (e - \alpha E) / (E - \alpha E)$ 
9:      $\mathcal{L} \leftarrow \text{Focal}(\gamma=2, w_c)$ 
10:  end if
11:  Draw mini-batches with sampling weights  $\propto n_c^{-\beta}$ 
12:  Update  $\theta$  by AdamW; clip gradient norm to 1
13:  Checkpoint if validation Macro-F1 improved
14: end for

```

 $(\bar{\mu}_c, \bar{\gamma}_c),$

$$\text{PPR}_c = \max\left(0, 1 - \frac{1}{2} \left[\frac{|\overline{\mu_{j,k}} - \bar{\mu}_c|}{\Delta_\mu} + \frac{|\overline{|\gamma_{j,k}|} - \bar{\gamma}_c|}{\Delta_\gamma} \right] \right), \quad (4)$$

where the overlines denote a mean of absolute per-edge deviations over all edges of group c , and $\Delta_\mu = 0.40$, $\Delta_\gamma = 0.20$ are fixed normalising spans chosen once as the plausible parameter ranges over all three components. The reported score is the unweighted mean of PPR_c over the three groups.

Two properties of (4) must be stated for the resulting number to be interpretable, and both were established empirically rather than assumed (§V-E). First, PPR is a *retention* measure: since PCWI initialises at the prior, an untrained PCWI model scores near-maximally by construction, so PPR can only decrease during training and a high value cannot show that the model *learned* physiological structure. Interpreting it requires the untrained value as a reference. Second, (4) averages each individual edge’s absolute deviation. Taking instead the deviation of each group’s *mean* parameter — the more obvious construction — is degenerate, because per-edge drift is close to symmetric and largely cancels: we measure a mean per-edge $|\Delta_\mu|$ of 0.064 against a group-mean $|\Delta_\mu|$ of 0.002, a factor of about 30. PPR is defined only for a PCWI-parameterised wavelet layer; it is not a general interpretability metric and does not apply to the comparators.

IV. DATA AND EVALUATION PROTOCOL

A. Preprocessing

MIT-BIH [22], [23] lead II is band-pass filtered per record, and a 360-sample window is extracted around each annotated R-peak. Each beat is z -scored using *its own* mean and standard deviation; no normalisation statistic is estimated across beats, records or splits, so no distributional information can cross the split boundary. Beats with degenerate variance are discarded. Inter-beat intervals are computed from annotation sample indices as described in §III-B.4.

B. Inter-Patient Split

We use the DS1/DS2 partition of de Chazal *et al.* [7] (Table I), splitting DS1 further into training and validation records following Takalo-Mattila *et al.* [1]. The partition is by record, so no patient contributes beats to more than one of the three subsets; disjointness is asserted programmatically and covered by a regression test. Checkpoint selection, early stopping and all architecture and hyperparameter decisions use DS1 validation only. DS2 is used to report the results of §V-B onward.

C. Class Distribution and AAMI Mapping

Annotations are mapped to the five AAMI EC57 super-classes [5], [24] (Table II). No class is excluded from the loss: every class receives the same inverse-frequency weight, with the single exception of the S-class multiplier of §III-C. The Q class is retained for completeness but there are only 15 Q beats in the entire database (7 in DS2), so no per-class conclusion can be drawn for it, and its contribution to any five-class macro average is dominated by sampling noise. We report Macro-F1 over all five classes for comparability with the protocol as conventionally applied, and report per-class recall separately so that this does not obscure where performance actually lies.

D. External Datasets

Cross-dataset evaluation uses the St Petersburg INCART 12-lead Arrhythmia Database [23] and the MIT-BIH Supraventricular Arrhythmia Database (SVDB) [25]. Both are processed by the identical pipeline described in §IV-A — same window length, same per-beat z -scoring, same interval formula, same clipping, same AAMI mapping — and both carry genuine expert per-beat annotations, so they are label-compatible with MIT-BIH at the granularity the model is trained on. SVDB was assembled specifically to enrich supraventricular ectopy and therefore functions as an S-class stress test. Neither dataset was used for any training, tuning or selection decision; both are evaluated forward-pass only from MIT-BIH-trained checkpoints.

We also processed PTB-XL [26] but **do not report it as a generalisation result**, because our own extraction pipeline for it is not distribution-compatible with the model’s training inputs in three independent ways, each sufficient to confound the measurement. (i) The beats are not z -scored, so the model receives raw millivolt amplitudes where (2)’s priors and the trained dilations $\gamma \approx 0.05$ – 0.12 are defined on the standardised scale; a narrow wavelet evaluated far from its translation contributes almost nothing, leaving the morphology branch not merely shifted but largely inactive. (ii) The interval features are divided by a local 10-beat running mean, making them dimensionless ratios near unity rather than the seconds the rhythm branch was trained on, which also compresses the prematurity contrast the branch depends on. (iii) PTB-XL’s SCP labels are record-level and our extraction applies one label to every beat of a record, so beat-level recall against them does not measure beat-level correctness. A Macro-F1 of

TABLE I
INTER-PATIENT DATA SPLITTING PROTOCOL (DS1/DS2).

Set	Role	Record IDs (MIT-BIH)	Function
DS1	Train (18 rec.)	101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 215, 220	Optimisation
	Val (4 rec.)	208, 209, 223, 230	All model and architecture selection
DS2	Test (22 rec.)	100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234	Held out

TABLE II
CLASS DISTRIBUTION AND AAMI STANDARD MAPPING.

Class	Included Annotations	Train	Val	Test (DS2)	Total	Ratio (vs N)
N	Normal, L, R, e, j	36,396	9,443	44,232	90,071	1.0 : 1
S	A, a, J, S	773	170	1,837	2,780	32.4 : 1
V	V, E	3,150	638	3,220	7,008	12.8 : 1
F	F	399	15	388	802	112.3 : 1
Q	/, f, Q	8	0	7	15	6004 : 1
Total		40,726	10,266	49,684	100,676	

0.200 $\pm$ 0.005 over 20 seeds was obtained, but it reflects an artefact of preprocessing rather than transfer, and we exclude it from all claims. Repeating the extraction under the pipeline of §IV-A and against beat-level annotations is the first item of §VI-E. We state the excluded result and its confounds explicitly, rather than simply omitting the dataset, so that the absence of a widely used benchmark is accounted for.

E. Baselines and What Is Matched

Four architectures spanning 29K–118K parameters are compared (Table III). Because the phrase “identical protocol” is often used loosely, we state exactly what is and is not held constant.

Held identical: dataset, preprocessing, DS1/DS2 split, input representation (both the beat window and the interval vector, so no model sees features another does not), optimiser (AdamW, learning rate 10^{-3} , weight decay 10^{-4}), cosine-annealed schedule, epoch budget (100), early-stopping patience (15), batch size (64), gradient-norm clipping (1.0), the checkpoint-selection rule (best DS1-validation Macro-F1), the set of 20 random seeds, and the class-weight formula.

Not held identical: the imbalance-handling mechanism, deliberately, since RAC is a contribution under test rather than a nuisance variable. The baselines use natural-distribution sampling with class-weighted cross-entropy, except CNN+Focal which uses focal loss without static class weights, matching each method’s own standard formulation. We tested applying one uniform imbalance mechanism to all five models and rejected it: stacking a class-balanced sampler with inverse-frequency weighting at this dataset’s imbalance ratio drives the majority class’s effective gradient contribution to near zero, and three of the four baselines then failed to predict class N at all. Degrading baselines with a mechanism chosen to suit the proposed model would not be a fair comparison. PC-WavKAN additionally applies minority-class signal augmentation; §V-H shows this does not account for its result, since removing it leaves Macro-F1 statistically unchanged.

TABLE III
BASELINE ARCHITECTURES. PROTOCOL, SPLIT, INPUT REPRESENTATION, OPTIMISER, BUDGET, CHECKPOINT RULE AND SEEDS ARE IDENTICAL TO PC-WavKAN (§IV-E); IMBALANCE HANDLING DIFFERS BY DESIGN.

Model	Params	Imbalance handling
ResNet1D (4-block residual CNN, 32 ch.)	62,869	Weighted CE
Transformer ($d_{\text{model}}=32$, 2 layers, 4 heads)	29,653	Weighted CE
CNN+Focal (4-layer 1D CNN)	71,061	Focal ($\gamma=2$)
B-Spline KAN (B-spline basis; no PCWI/P-region)	118,229	Weighted CE
PC-WavKAN (Mexican Hat KAN, PCWI, gated P-region fusion, MLP rhythm enc.)	153,045	RAC (§III-C)

Table V should therefore be read as comparing complete methods — architecture together with its imbalance strategy — under a common protocol and budget, not as isolating the architecture alone; the within-family ablation of §V-A is what isolates architectural components.

F. Statistical Methodology

Every arm is trained with the same 20 random seeds, and every comparison is paired by seed: seed s of one arm is compared with seed s of the other, so the pairing corresponds to a shared initialisation and data-order stream. Twenty seeds were chosen to make a medium standardised effect detectable at conventional power under a non-parametric paired test while keeping the study’s roughly 400 training runs tractable; the count was fixed before any comparison was run, and no arm was later extended or truncated.

Significance is assessed by the two-sided Wilcoxon signed-rank test [27], which does not assume normality of the per-seed differences. Because the study reports many metrics, we declare comparison families explicitly and apply Holm–Bonferroni correction [28] within each: the metrics of a single arm-vs-arm comparison form one family (five metrics for a per-class comparison, four where fusion recall is not estimable), and the set of comparisons against the four baselines on one metric forms one family. All p -values reported are

Holm-adjusted within the stated family unless labelled raw. Effect sizes are Cohen’s d [29] for paired differences, $d = \bar{\delta}/s_{\delta}$, and we adopt one sign convention throughout: **positive d favours PC-WavKAN, the adopted configuration, or the treatment arm.** We additionally report 95% confidence intervals on the paired mean difference, because for the primary comparison the width of that interval is more informative than the non-significance of the test.

We distinguish three levels of claim. The *primary* outcome is Macro-F1 on DS2 for the adopted configuration against the four baselines (§V-B). *Secondary*, pre-specified outcomes are the per-class recalls, on which the curriculum claim rests (§V-C). Everything else — the rhythm-position ablation, the cross-dataset confusion analysis, the few-shot curves and the augmentation comparison — is *exploratory*: hypothesis-generating, corrected within its own family where a test is reported, and not to be read as confirmatory.

V. RESULTS

A. Architecture Selection on Validation Data

The configuration reported in the rest of this paper was chosen using DS1 validation only. Table IV gives the single-variable ablation: the base configuration is the wavelet KAN with PCWI, gated P-region fusion, the self-attention rhythm encoder and RAC, and each row changes exactly one component of that base. All 20 seeds are trained for every row, and the validation figure for each seed is taken at the epoch its own checkpoint was saved, so no additional training or test-set access was involved.

Three findings follow, and they revise the account of the architecture its initial design implied.

PCWI is the architecture’s effective novel component. Replacing (2) with isotropic random initialisation costs 0.021 validation Macro-F1 with a large effect ($d = -1.05$, Holm $p = 1 \times 10^{-4}$). This is the only positive contribution among the components we introduced that survives correction. Because every row of Table IV varies one component of the *base* configuration, this establishes PCWI’s contribution within that base; the adopted configuration differs from it only in the rhythm encoder, and the two components act on different branches of the network, so we expect the effect to carry over — but we have not measured it there, and we do not present it as though we had. Re-running this single ablation on the adopted configuration is a well-defined experiment requiring no new methodology, and is the study’s main outstanding item (§VI-E).

The mother wavelet is interchangeable. Substituting Morlet, DOG or a B-spline for the Mexican Hat changes validation Macro-F1 by at most 0.010, and none of the three differences approaches significance; Morlet is in fact numerically higher than the base. The Mexican Hat is a reasonable choice on the standard argument that its shape matches a sharp biphasic transient [11], but our own data do not support a claim that this choice is what makes the model work. What matters measurably is *where the basis is initialised*, not *which* basis it is — which is consistent with PCWI being the component that carries the effect, and is a more specific conclusion than a general appeal to wavelet suitability.

The self-attention rhythm encoder is harmful and is removed. Replacing it with a plain MLP over the same five intervals improves validation Macro-F1 by 0.025 with a large effect ($d = +1.15$, Holm $p = 8 \times 10^{-4}$), and is the best of all seven configurations on validation. We adopt this 153,045-parameter configuration. §VI-B discusses why self-attention over five scalars is a poor inductive bias here.

Disclosure regarding test-set exposure: Table IV is a validation-only analysis and its ranking is sufficient to select the adopted configuration without reference to DS2. We note for completeness that DS2 metrics for the non-adopted variants were also computed during development, and that the DS2 ranking agrees with the validation ranking (the adopted configuration is also highest on DS2, at 0.357 ± 0.019 against 0.323 ± 0.018 for the base). DS2 was therefore not literally untouched over the whole course of this study, and we do not describe it as such. What the design does guarantee is that the selection rule is fully determined by validation data, that no hyperparameter, threshold or checkpoint was ever chosen on DS2, and that the primary comparison of §V-B compares one pre-selected configuration against four baselines subjected to the same selection rule. Readers who prefer the most conservative reading may treat §V-B as a held-out estimate under a validation-determined selection procedure rather than as a single-shot test result; the conclusion of that section is a null result and is not the kind of finding that selection bias would manufacture.

B. Primary Comparison Against Standard Architectures

Table V reports the primary outcome: DS2 Macro-F1 for the adopted configuration against the four baselines, 20 paired seeds each, analysed as specified in §IV-F.

PC-WavKAN’s Macro-F1 is statistically indistinguishable from all four baselines. All Holm-adjusted $p \geq 0.36$, all effect sizes are negligible-to-small, and — more informative than the p -values — every 95% confidence interval on the paired difference contains zero, with the widest bound 0.029 in absolute Macro-F1. The data therefore exclude any large advantage in either direction, and we make no accuracy claim beyond parity. We emphasise that this is a null result at $n = 20$ per arm, not a demonstration of equivalence: the intervals are compatible with differences of up to roughly ± 0.03 Macro-F1, and a formal equivalence claim would require a pre-specified margin, which we did not set.

Two features of this comparison deserve note. It is not parameter-favourable to PC-WavKAN, which is the largest of the five models and yet does not outperform a Transformer with a fifth of its parameters. And the absolute values are low for all five models: Macro-F1 near 0.36 under this protocol reflects the S- and F-class recalls that the inter-patient setting collapses, not a defect specific to one architecture. Fig. 3 shows the structure behind this for PC-WavKAN. N and V separate well, each leaking under 6% into the other, and V-recall is 0.905. The Macro-F1 ceiling is set by the other classes: S beats are misassigned predominantly to V (0.42) and secondarily to N (0.30), and F beats are almost entirely predicted as N (0.87). The S→V direction is the more diagnostically interesting of the two and recurs in the cross-dataset analysis of §V-F.

TABLE IV

SINGLE-VARIABLE COMPONENT ABLATION ON **DS1 VALIDATION**, 20 SEEDS PER ROW. BASE CONFIGURATION: PCWI + GATED P-REGION FUSION + SELF-ATTENTION RHYTHM ENCODER + RAC, VALIDATION MACRO-F1 0.4145 ± 0.0201 . Δ AND d ARE VARIANT MINUS BASE, SO POSITIVE FAVOURS THE VARIANT. p IS HOLM-ADJUSTED ACROSS THE SEVEN COMPARISONS. ONLY TWO COMPONENTS HAVE A CORRECTED EFFECT: REMOVING PCWI HURTS, AND REMOVING THE SELF-ATTENTION RHYTHM ENCODER HELPS.

Variant	Val. Macro-F1	Δ vs base	Cohen's d	Holm p	Interpretation
No PCWI (random init.)	0.3932 ± 0.0129	-0.0213	-1.05 (large)	0.0001	PCWI contributes
No gated P-region fusion	0.3976 ± 0.0184	-0.0169	-0.54 (medium)	0.164	not resolved
No RAC (natural sampling)	0.4053 ± 0.0173	-0.0092	-0.37 (small)	0.624	not resolved on Macro-F1
Basis: Morlet	0.4219 ± 0.0187	$+0.0074$	$+0.31$ (small)	0.568	basis interchangeable
Basis: DOG	0.4041 ± 0.0123	-0.0104	-0.45 (small)	0.164	basis interchangeable
Basis: B-spline	0.4116 ± 0.0151	-0.0029	-0.14 (negl.)	0.624	basis interchangeable
No self-attn. rhythm encoder	0.4397 ± 0.0140	$+0.0252$	$+1.15$ (large)	0.0008	adopted

TABLE V

PRIMARY OUTCOME. DS2 MACRO-F1, PC-WavKAN VS. FOUR BASELINES, 20 PAIRED SEEDS. Δ , CI AND d ARE PC-WavKAN MINUS BASELINE; p IS HOLM-ADJUSTED ACROSS THE FOUR COMPARISONS. NO COMPARISON REACHES SIGNIFICANCE AND EVERY CI INCLUDES ZERO.

Model	Params	Macro-F1	Δ	95% CI of Δ	d / Holm p
ResNet1D	62,869	0.351 ± 0.019	$+0.006$	$[-0.008, +0.020]$	$+0.19$ / 1.00
Transformer	29,653	0.353 ± 0.040	$+0.004$	$[-0.013, +0.022]$	$+0.12$ / 1.00
CNN+Focal	71,061	0.362 ± 0.015	-0.005	$[-0.017, +0.006]$	-0.22 / 1.00
B-Spline KAN	118,229	0.371 ± 0.024	-0.014	$[-0.029, +0.001]$	-0.43 / 0.36
PC-WavKAN	153,045	0.357 ± 0.019	—	—	—

TABLE VI

SECONDARY OUTCOME. RAC VS. NATURAL-DISTRIBUTION SAMPLING, PC-WavKAN, 20 PAIRED SEEDS, DS2. AUGMENTATION DISABLED IN BOTH ARMS. Δ AND d ARE RAC MINUS NO-RAC; p IS HOLM-ADJUSTED ACROSS THE FIVE METRICS.

Metric	No RAC	RAC	Cohen's d	Holm p
Macro-F1	0.351 ± 0.014	0.364 ± 0.021	$+0.52$ (med.)	0.098
S-Recall	0.137 ± 0.037	0.222 ± 0.057	$+1.47$ (large)	2.9×10^{-5}
F-Recall	0.0003	0.0022	$+0.88$ (large)	0.013
V-Recall	0.880 ± 0.024	0.880 ± 0.017	-0.00 (negl.)	0.756
N-Recall	0.927 ± 0.021	0.917 ± 0.024	-0.32 (small)	0.228

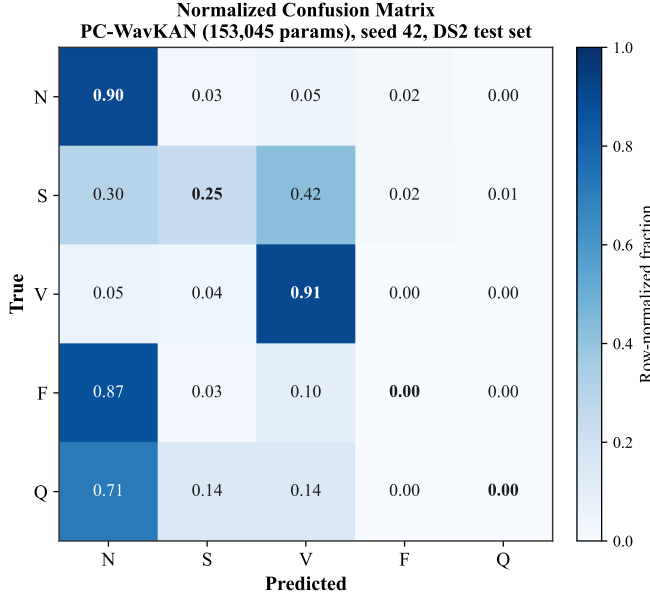


Fig. 3. Row-normalised confusion matrix, PC-WavKAN, seed 42, DS2. Generated from this checkpoint's own predictions.

C. Effect of the Rarity-Anchored Curriculum

Table VI isolates RAC on the adopted architecture. Both arms are the same 153,045-parameter model with the same optimiser, budget and seeds, and augmentation is disabled in both, so the schedule of §III-C is the only variable.

RAC produces a large increase in S-recall, from 0.137 to 0.222 ($d = +1.47$ after correction across five metrics), with a 95% confidence interval on the paired difference of $[+0.058, +0.112]$, well clear of zero. Fusion recall also

improves significantly in relative terms ($d = +0.88$, CI $[+0.0009, +0.0030]$), though it remains negligible in absolute terms and we draw no practical conclusion from a recall of 0.002. V-recall is unchanged and N-recall's small decrease does not survive correction. Macro-F1 trends upward with a medium effect ($+0.012$, CI $[+0.001, +0.023]$) but does not reach significance — expected, since a five-class macro average dilutes a gain confined to one minority class.

The defensible statement is therefore narrow and we keep it narrow: RAC improves supraventricular recall substantially and reliably, at no statistically robust cost to any other class, and it does not by itself materially move the aggregate metric. An S-recall of 0.222 still misses roughly four in five supraventricular beats; this is a real improvement on a hard class, not a solution to it.

The same effect replicates on the base configuration with the self-attention rhythm encoder and with augmentation enabled, where S-recall rises from 0.079 ± 0.016 to 0.123 ± 0.052 (Holm $p = 0.0016$, $d = +0.77$, medium), and on DS1 validation, where it rises from 0.218 ± 0.048 to 0.305 ± 0.047 ($d = +1.36$, raw $p = 2.4 \times 10^{-4}$). The effect is thus reproducible across two architectural variants, two training recipes and both data partitions, which is the basis on which we treat it as a property of the schedule rather than of a particular run. Fig. 4 shows the per-epoch dynamics for a representative seed.

D. Dependence on Rhythm Features and Non-Causality

Table VII reports an exploratory leave-one-out analysis over the five interval positions: each position is masked at inference and the change in S-recall measured across 20 seeds. Because the interval feature set is identical in both rhythm encoders, this analysis was run on the base configuration,

PC-WavKAN training dynamics: no-RAC vs. RAC (seed 42)
self-attention rhythm encoder, 154,325 params; 22 vs. 17 epochs (independent early stopping, patience 15)

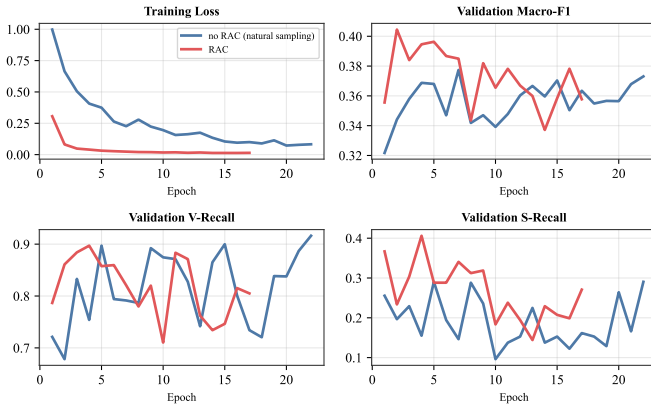


Fig. 4. Per-epoch training dynamics, no-RAC vs. RAC arm (seed 42, DS1 validation Macro-F1), on the base configuration used for the replication in §V-C.

TABLE VII

LEAVE-ONE-OUT INTERVAL ABLATION, 20 SEEDS, DS2, EXPLORATORY. BASELINE S-RECALL WITH THE FULL VECTOR: 0.1225 ± 0.0524 . p IS HOLM-ADJUSTED ACROSS THE FIVE POSITIONS.

Masked position	Δ S-Recall	Holm p	Sig.?
RR_{-2}	-0.0020 ± 0.0077	0.221	No
RR_{-1}	-0.0016 ± 0.0075	0.221	No
RR_0 (pre-beat)	-0.0323 ± 0.0492	3×10^{-4}	Yes
RR_{+1} (post-beat)	$+0.0022 \pm 0.0095$	0.358	No
RR_{+2} (post-beat)	-0.0045 ± 0.0103	0.014	Yes (small)

which is the variant for which the 20 masked-inference passes were available; it characterises the feature set rather than the encoder.

Masking RR_0 , the interval immediately preceding the classified beat, causes by far the largest degradation. This is the expected result and we present it as a sanity check rather than a discovery: a shortened pre-beat interval is the textbook signature of a premature ectopic beat, so it should be the single most important element of the vector, and the fact that the model’s dependence structure recovers this is evidence that the rhythm branch is using the feature the way clinical reasoning would.

The consequential row is RR_{+2} . Its effect is small but survives correction, which means the model draws real diagnostic signal from an interval that has not yet elapsed when beat i would need to be labelled in a streaming deployment. Together with the record-level filtering noted in §III, this establishes the non-causality of §III as a substantive property rather than a formality: simply zeroing the two post-beat positions at inference would be expected to cost accuracy, not merely to remove an unused input. We report the corresponding V-recall change at RR_0 (-0.0745 ± 0.1132) as a raw effect only, since we did not include V-recall in this analysis’s test family.

E. Retention of the Physiological Prior

Table VIII evaluates PPR (§III-D, (4)) on all 20 trained checkpoints against the two reference points that make the

TABLE VIII

PHYSIOLOGICAL PRIOR RETENTION (4), MEAN $\pm$ STD OVER 20 MODELS, WITH THE MEAN ABSOLUTE PER-EDGE PARAMETER DEVIATION FROM THE PRIOR. THE UNTRAINED PCWI ROW IS THE NULL: PCWI INITIALISES AT THE PRIOR (UP TO ITS SYMMETRY-BREAKING JITTER), SO IT SCORES NEAR-MAXIMALLY BY CONSTRUCTION AND PPR CAN ONLY DECREASE WITH TRAINING.

Model state	PPR	$ \Delta\mu $	$ \Delta\gamma $
Untrained, PCWI (<i>null</i>)	0.9729 ± 0.0001	0.013	0.004
Trained, PCWI (PC-WavKAN)	0.7826 ± 0.0351	0.064	0.055
Untrained, random init. (<i>floor</i>)	0.6698 ± 0.0008	0.161	0.051

number interpretable: the same architecture untrained with PCWI active, and untrained with PCWI replaced by isotropic random initialisation.

The result is best stated as what it does and does not support. Training moves the wavelet parameters substantially: PPR falls from its near-maximal initial value of 0.973 to 0.783, and the mean absolute per-edge translation deviation grows from the 0.013 contributed by PCWI’s own symmetry-breaking jitter to 0.064, a factor of five, on a normalising span of 0.40. The trained basis nonetheless remains far closer to the ECG priors than a randomly initialised basis of the same architecture (0.783 vs 0.698; Mann–Whitney U , $p = 3 \times 10^{-8}$, standardised difference +4.6), and the gap is highly consistent across seeds. The QRS group retains the most structure (PPR = 0.814), the P and T groups slightly less (0.764, 0.770). This ordering is by absolute parameter deviation and is deliberately not an ordering by visual similarity of the resulting functions: in Fig. 5 the P group’s curves appear to sit almost on their prior while the QRS group’s are visibly wider, because a wavelet initialised narrow ($\bar{\gamma} = 0.05$) changes shape far more conspicuously than a wide one ($\bar{\gamma} = 0.12$) for the same absolute change in γ . PPR measures the parameters, and we report it that way rather than substituting an impression of the plotted curves.

So the defensible claim is that the physiologically structured function space established by (2) is *retained*, not discarded, under gradient descent, while still being refined by it — and §V-A independently shows this initialisation is the component that measurably improves validation performance. The two findings together support a specific and modest interpretation: constraining where the basis starts is useful, and the useful structure persists. They do not support the stronger claim that the model learned physiological alignment, which the null in Table VIII directly refutes, since a high PPR is available to an untrained network. Nor is PPR a measure of interpretability in the sense of explaining a prediction: it is computed entirely from parameters and never observes a model output, so it says nothing about whether an individual classification is attributable to the component a channel group is named for. We report it as a transparency property of the parameterisation, and we regard publishing its null as inseparable from reporting the score at all. Fig. 5 shows the trained bases against their priors.

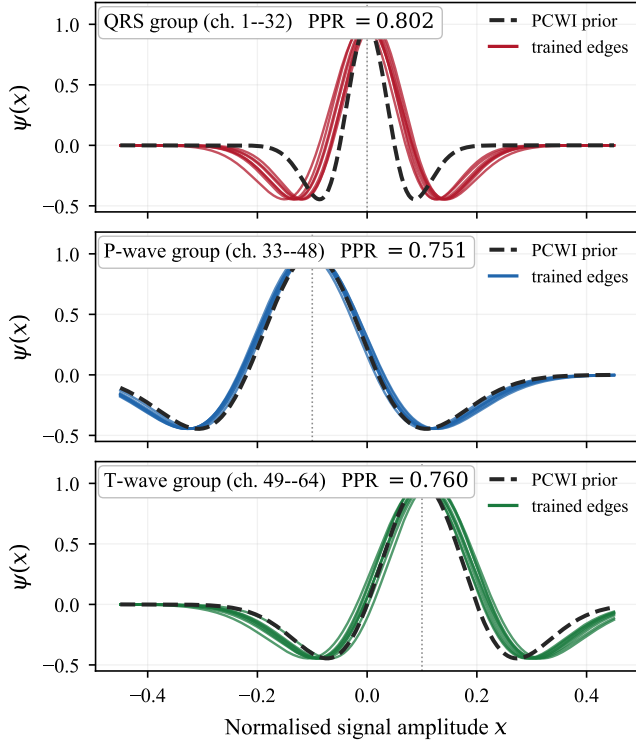


Fig. 5. **Trained wavelet bases per channel group**, against the PCWI prior each group was initialised from (dashed). Curves are read from one checkpoint (seed 42), so annotated retention differs slightly from Table VIII’s 20-seed means. Parameters move from initialisation while the group structure of (2) persists.

F. Cross-Dataset Generalisation

Table IX evaluates all five MIT-BIH-trained models on INCART and SVDB, forward-pass only, 20 seeds each. Both datasets pass through the identical pipeline as the training data (§IV-D), so these are dataset-shift measurements rather than preprocessing artefacts.

The pattern is dataset-dependent and we do not summarise it as a single verdict. On INCART, PC-WavKAN has the highest Macro-F1 of the five models and is significantly better than three of the four baselines with large effects, but it is *not* significantly better than B-Spline KAN ($d = +0.45$, Holm $p = 0.083$); the honest reading is that the two KAN models lead this dataset and the numerical ordering between them is unresolved. On SVDB, PC-WavKAN is significantly worse than the same three baselines, again with large effects, and again statistically tied with B-Spline KAN.

That the two KAN models track each other on both datasets — leading together on one, trailing together on the other — points to shared KAN structure rather than anything specific to PC-WavKAN. A per-class analysis over summed confusion matrices localises the mechanism. On SVDB both KAN models misassign supraventricular beats to V at a nearly identical, elevated rate (PC-WavKAN 0.408, B-Spline KAN 0.412) that the non-KAN baselines do not approach (ResNet1D 0.173, CNN+Focal 0.118, Transformer 0.319). Both also over-predict V there (17.8% and 18.9% of all predictions, against 8.3% and 5.5% for ResNet1D and CNN+Focal) and correspondingly

attain the two highest V-recalls of the five models. The bias that helps ventricular detection is what costs supraventricular detection, and on a dataset enriched for supraventricular ectopy that trade is unfavourable. The same signature is far weaker on INCART, where PC-WavKAN’s S→V rate (0.280) exceeds the highest non-KAN baseline (0.253) by only 0.027, against a gap of 0.089 on SVDB. This localises the failure mode — a systematic S→V confusion tied to V over-prediction, shared across the wavelet- and B-spline-basis KANs, and expressed most strongly where supraventricular beats are most prevalent — without yet explaining why the KAN parameterisation induces it (§VI-E).

G. Few-Shot Domain Adaptation

The results above are zero-shot. We next ask, as an exploratory analysis, how much a small amount of labelled target-domain data changes them. For each dataset we fine-tune from the zero-shot checkpoint on $k \in \{0, 10, 50, 100, 500\}$ beats per class, drawn from a record-disjoint adaptation pool fixed by a constant split seed before any model sees the data and held out from the evaluation pool used for every number reported. Each k is fine-tuned independently from a fresh copy of the original checkpoint, and where a class has fewer than k beats we use the real smaller count rather than resampling (this affects only F and Q shots at $k \geq 50$). As a consistency check, the $k = 0$ recomputation on the held-out pool lands close to Table IX (INCART 0.392 vs 0.374; SVDB 0.271 vs 0.281), the difference reflecting the smaller pool.

Table X shows adaptation is not a general fix: it reallocates recall between classes, and whether that helps the aggregate is dataset-dependent. On SVDB it works clearly — S-recall rises almost fivefold ($0.107 \rightarrow 0.506$, $d = +7.07$) and Macro-F1 improves significantly ($0.271 \rightarrow 0.331$, $d = +4.58$) even as V-recall falls. On INCART Macro-F1 does not change significantly: V-recall improves substantially ($d = +2.96$) but S-recall *halves* ($0.571 \rightarrow 0.295$, $d = -1.90$), and the two cancel in the macro average. The curve is also non-monotonic, which we report rather than reduce to its endpoint: INCART’s S-recall peaks at $k = 50$ (0.643 ± 0.095) and falls below its own zero-shot value by $k = 500$. Choosing k for this architecture therefore requires the whole curve and the class of interest, not a single favourable operating point.

H. Augmentation

Because PC-WavKAN’s training recipe applies minority-class signal augmentation while the baselines do not (§IV-E), we quantify what that asymmetry is worth. Table XI compares four augmentation strategies against none, 20 seeds each, under an isolated recipe (60 epochs, no RAC) chosen so that augmentation is the only active mechanism; its baseline is therefore not comparable to Table VI.

Under this isolated recipe all three noise-based strategies significantly improve V-recall (Holm $p < 0.002$, $d \geq +1.16$) and significantly *reduce* Macro-F1 (Holm $p \leq 0.033$, $d \in [-1.08, -0.50]$), while none reaches significance on S-recall

TABLE IX

ZERO-SHOT CROSS-DATASET EVALUATION, 20 SEEDS PER MODEL. SIGNIFICANCE IS PC-WavKAN VS. EACH BASELINE ON MACRO-F1, HOLM-ADJUSTED WITHIN EACH DATASET; POSITIVE d FAVOURS PC-WavKAN. BOLDFACE MARKS THE NUMERICALLY HIGHEST VALUE PER COLUMN, WHICH FOR MACRO-F1 ON INCART IS NOT A SIGNIFICANT WIN OVER B-SPLINE KAN.

Model	INCART				SVDB (S-class stress test)			
	Macro-F1	d / Holm p	V-Rec	S-Rec	Macro-F1	d / Holm p	V-Rec	S-Rec
PC-WavKAN	0.374 ± 0.009	—	0.709	0.500	0.281 ± 0.013	—	0.794	0.135
ResNet1D	0.344 ± 0.018	+1.62 / < 0.001	0.489	0.505	0.370 ± 0.023	−3.26 / < 0.001	0.733	0.324
Transformer	0.335 ± 0.031	+1.27 / < 0.001	0.626	0.682	0.335 ± 0.035	−1.52 / < 0.001	0.786	0.377
CNN+Focal	0.316 ± 0.024	+2.02 / < 0.001	0.346	0.449	0.367 ± 0.013	−4.29 / < 0.001	0.642	0.247
B-Spline KAN	0.365 ± 0.017	+0.45 / 0.083	0.732	0.386	0.276 ± 0.018	+0.19 / 0.648	0.789	0.125

TABLE X

FEW-SHOT ADAPTATION, PC-WavKAN, 20 SEEDS PER CELL, EXPLORATORY. FULL CURVES FOR $k = 10, 50, 100$ ARE IN THE RELEASED RESULTS; * MARKS HOLM-ADJUSTED $p < 0.05$ WITHIN EACH DATASET'S FOUR-METRIC FAMILY.

Dataset	k	Macro-F1	V-Rec	S-Rec	N-Rec
INCART	0	0.392	0.726	0.571	0.954
	500	0.396	0.850*	0.295*	0.958
SVDB	0	0.271	0.774	0.107	0.788
	500	0.331*	0.707*	0.506*	0.782

TABLE XI

AUGMENTATION STRATEGIES, PC-WavKAN, 20 SEEDS, DS2, ISOLATED RECIPE (60 EPOCHS, NO RAC), EXPLORATORY. p IS HOLM-ADJUSTED ACROSS THE FOUR AUGMENTED ARMS.

Strategy	Macro-F1	V-Recall	S-Recall
None	0.336 ± 0.031	0.884 ± 0.017	0.316 ± 0.050
Gaussian noise	0.303 ± 0.010	0.914 ± 0.016	0.333 ± 0.038
Baseline wander	0.320 ± 0.009	0.917 ± 0.010	0.339 ± 0.051
Combined	0.293 ± 0.018	0.922 ± 0.030	0.337 ± 0.048
SMOTE	0.201 ± 0.026	0.869 ± 0.062	0.648 ± 0.105

($p \geq 0.16$). SMOTE is the clearest trade: the largest S-recall gain measured anywhere in this study ($0.316 \rightarrow 0.648$, $d = +3.01$) at the largest Macro-F1 cost ($d = -4.58$), by interpolating beats that push the decision boundary into the S region at every other class's expense. We do not recommend it, consistent with Bae and Shin [17].

This Macro-F1 cost does *not* carry over to the full recipe, which is what matters for interpreting Table V. Tested under the actual training configuration (100 epochs, RAC, 20 seeds), removing augmentation leaves Macro-F1 statistically unchanged ($0.357 \rightarrow 0.364$, Holm $p = 0.46$, $d = +0.25$) while significantly reducing F-recall ($0.0061 \rightarrow 0.0022$, Holm $p = 0.043$) and trending toward lower V-recall ($d = -0.70$, Holm $p = 0.083$). Two conclusions follow. The augmentation asymmetry with the baselines does not explain PC-WavKAN's parity result, since removing it does not reduce Macro-F1. And the isolated recipe's penalty is masked once RAC and the full budget are present, so short single-mechanism ablations of this kind should not be extrapolated to full recipes; which factor is responsible is left open (§VI-E).

TABLE XII

MEASURED DEPLOYMENT METRICS, PC-WavKAN, x86 CPU, SINGLE THREAD, PYTORCH 2.6. MEAN OF 3 REPEATED RUNS; $\pm$ IS THE SPREAD ACROSS RUNS.

Metric	PC-WavKAN (153,045 params)
Model size, FP32	0.596 MB
Model size, INT8 dynamic	0.551 MB
Size reduction	1.08×
Latency, batch 1, FP32	2.93 ± 0.38 ms/beat (median 2.78)
Latency, batch 1, INT8	4.00 ± 0.35 ms/beat
INT8 vs. FP32 speedup	$0.73 \times$ (<i>slower</i>)
Latency, batch 64, FP32	0.429 ms/beat
Peak resident memory	235 MB

I. Efficiency and Quantisation

Table XII reports measured inference behaviour on x86 CPU, single-threaded, for the seed-42 checkpoint. Each figure is the mean of three independently repeated runs of 1000 timed trials after a 20-iteration warm-up: a single run can carry a heavy right tail from transient host load (we observed one with a threefold mean-to-median gap that vanished on repetition). We therefore also report the median, and quote the spread *across* runs rather than the within-run per-trial standard deviation, which describes the timing tail rather than the uncertainty of the estimate.

Post-training dynamic INT8 quantisation gives essentially nothing here and makes inference slower. This is explicable rather than anomalous: PyTorch's dynamic quantisation converts `nn.Linear` modules only, whereas this model's parameters sit almost entirely in the custom wavelet-KAN layer and the bidirectional GRU, so the pass adds dequantisation overhead without accelerating the dominant part of the graph. An efficiency claim for a KAN-based architecture should therefore rest on measured FP32 latency and parameter count, not an assumed quantisation benefit; realising one would require a quantised wavelet-layer implementation.

At 2.93 ms per beat single-threaded and 0.429 ms batched, throughput far exceeds the 1–2 beats per second of a resting adult, so the model is compatible with continuous processing on a general-purpose CPU. That is not evidence of wearable deployability: these are x86 measurements, and we have not benchmarked on the ARM Cortex-class microcontrollers on-device Holter analysis would use, where the absent quantisation path and the GRU's sequential dependency are both likely to matter more. We therefore describe the model as resource-

efficient with measured x86 CPU inference, and no more, and we do not tabulate it against MAK-Net’s reported 24 ms/beat GPU figure [12]: the hardware classes differ and no conclusion should be drawn in either direction.

VI. DISCUSSION

A. What This Study Establishes

Three findings are supported at the level of rigour the protocol was built for. First, physiology-constrained initialisation of a wavelet-KAN basis measurably improves validation performance within the configuration it was ablated in, is the only novel component of the architecture with a corrected large effect there, and its structure is substantially retained through training on the adopted model. Second, a rarity-anchored curriculum produces a large, reproducible improvement in supraventricular recall at no robust cost elsewhere, replicated across two architectural variants, two recipes and both partitions. Third, under a matched protocol with multiplicity control, this architecture is statistically indistinguishable from four standard baselines on Macro-F1, with all paired confidence intervals containing zero.

The third finding constrains how the first two should be read. PC-WavKAN is not more accurate than a 30K-parameter Transformer here, and we do not present the wavelet-KAN basis as an accuracy mechanism. What it offers is a parameterisation in which the learned function space is expressible in morphological terms, can be initialised from physiological knowledge, and can be measured against that knowledge after training — obtained, on this evidence, at no accuracy cost relative to standard architectures. Whether that is worth the parameter budget depends on how much an application values inspectability of the basis; that is an argument about design priorities, and we keep it separate from the accuracy evidence. On protocol rather than performance, the study satisfies all four criteria of Silva *et al.*’s E3C framework [6] — inter-patient evaluation, class-specific metrics under imbalance, AAMI conformance and consideration of computational cost — which is compatible with, and independent of, the parity finding.

An unexpected consequence of the ablation is that it narrows our own architectural claim. The specific mother wavelet turns out not to matter measurably (Table IV), so the appeal to the Mexican Hat’s suitability for sharp QRS transients [11] — which motivated the design — is not what the data support. The transferable insight is about *initialisation* in a parameterisation whose parameters have physiological units, and it should transfer to any wavelet basis, which is a broader claim than one tied to a particular function.

B. Why Self-Attention Over Interval Features Fails

Removing self-attention from the rhythm encoder improved validation Macro-F1 with a large effect (Table IV), and this is not in tension with §V-D’s finding that the interval features carry real signal. The two results together indicate that the problem is the encoder, not the features.

Self-attention computes content-based pairwise interactions over a set and is, by construction, close to permutation-equivariant absent positional information. That is well matched to long sequences whose informative structure is relational. The interval vector is the opposite case: it has five elements, and §V-D shows its information is sharply concentrated in one *position* — RR_0 , the pre-beat interval — whose diagnostic meaning derives entirely from which position it occupies. An encoder whose inductive bias is order-insensitive relational mixing must expend capacity relearning the positional distinction that a plain MLP over the fixed-length vector has for free, on 20 seeds’ worth of a minority class. The negative result is therefore informative rather than incidental: it suggests attention is the wrong primitive for short fixed-length physiological feature vectors with strong positional semantics, and that this should be checked rather than assumed whenever an attention block is added to such a branch.

C. The Cross-Dataset Pattern

The INCART and SVDB results (§V-F) are not a single generalisation verdict, and the informative structure is that the two KAN models move together: both lead on INCART, both trail on SVDB, and the per-class analysis attributes the SVDB deficit to an $S \rightarrow V$ confusion tied to V over-prediction that both share and the non-KAN baselines do not. Since the signature is common to a wavelet and a B-spline basis, it is unlikely to be specific to the wavelet parameterisation. The more plausible reading is that the KAN edge parameterisation, in which each input sample is transformed by its own univariate function before summation, favours localised amplitude-morphology evidence over the broader contextual cues that separate supraventricular from ventricular ectopy when the two are morphologically close, as in SVDB’s population. This is a hypothesis consistent with the diagnostic rather than a demonstrated mechanism, and is testable by ablating the KAN layer’s locality with the rest of the architecture fixed.

The finding also cautions against reporting a single external dataset: on INCART alone PC-WavKAN would appear the best of five models at transfer, on SVDB alone among the worst, and either conclusion would have been backed by 20 seeds and corrected statistics.

D. Limitations

(1) **Parity, not advantage.** Macro-F1 is statistically indistinguishable from four baselines (§V-B); a reader expecting an accuracy gain from a new architecture will not find one. The intervals bound any true difference at roughly ± 0.03 but do not establish equivalence, for which we set no margin.

(2) **Absolute minority-class performance.** F-recall is negligible (≈ 0.006) for all configurations, reflecting 388 test beats and 399 training beats; the significant relative improvement in Table VI has no practical clinical reading at that absolute level. Q, with 15 beats in the database, supports no per-class conclusion at all and inflates the variance of any five-class macro average. S-recall, at 0.222, still misses about four in five supraventricular beats.

(3) Selection history of the test set. The adopted configuration is fully determined by validation data (§V-A), but DS2 metrics for non-adopted variants were computed during development, so DS2 is not a never-inspected partition and we do not describe it as one.

(4) Non-causal inputs. The interval vector includes two post-beat intervals and the filter is applied per record (§III), so the model performs offline classification and would need a two-beat lookahead buffer and a causal filter to run in a streaming setting. §V-D shows the lookahead carries significant signal, so the cost of removing it is real and, as yet, unquantified.

(5) SVDB weakness. PC-WavKAN is significantly below three of four baselines on SVDB (§V-F). The failure mode is localised to an S→V bias shared with B-Spline KAN, but its cause in the KAN parameterisation is a hypothesis, not a result.

(6) PTB-XL not evaluated. Our PTB-XL extraction is confounded by three independent preprocessing and labelling mismatches (§IV-D), so we report no transfer result for it. Cross-dataset evidence in this paper rests on two external databases, both of which share MIT-BIH’s annotation granularity and were processed identically; generalisation to 12-lead clinical recordings with record-level diagnostic labels is untested here.

(7) Adaptation trades classes rather than closing gaps. Few-shot adaptation improves the classes it is given data for at other classes’ expense, non-monotonically in k on INCART (§V-G); we have not characterised which target-domain distributions make the trade favourable.

(8) Hardware scope. Latency is x86 CPU only; no ARM-class or on-device measurement was made, and dynamic INT8 quantisation does not benefit this architecture (§V-I).

(9) The component ablation is on the base configuration, not the adopted one. Every row of Table IV varies one component of the base configuration, which is the correct design for the selection that table performs, but it means PCWI — the component this paper’s architectural contribution rests on — has been ablated only there and not on the model we finally adopt. The two differ solely in the rhythm encoder, and PCWI acts on the morphology branch, so we expect the effect to transfer; that expectation is reasoning, not measurement, and we mark it as the study’s principal outstanding experiment (§VI-E) rather than assume it. The same applies to the P-region pathway and the basis comparison, whose effects were in any case not resolved after correction.

E. Future Work

The first item is to re-run the PCWI ablation on the adopted configuration (§VI-D, item 9): 20 seeds, one variant, no new methodology, and it converts the paper’s central architectural claim from one established on the base configuration into one measured on the model actually proposed. Second is to repeat the PTB-XL evaluation under the pipeline of §IV-A and against beat-level annotations, which would convert an excluded, confounded result into a real test of transfer to 12-lead clinical recordings. Beyond those we prioritise: identifying the mechanism behind the shared KAN S→V bias by ablating the edge parameterisation’s locality while holding

the rest of the architecture fixed; strengthening the prior-retention analysis from a parameter-space measurement into one linked to model outputs, for instance by testing whether perturbing a channel group’s wavelets selectively degrades the class whose morphology that group is initialised for, which would let a transparency claim rest on behaviour rather than on parameters alone; retraining with a strictly causal interval vector and a causal filter to quantify what streaming compatibility costs; evaluating adaptive logit adjustment [19] as a beat-level alternative to RAC; and benchmarking on ARM-class hardware with a quantised wavelet-layer implementation.

VII. CONCLUSION

We presented PC-WavKAN, a 153,045-parameter wavelet Kolmogorov–Arnold network for inter-patient ECG beat classification whose distinguishing component is physiology-constrained initialisation of the wavelet basis. Evaluated under the de Chazal DS1/DS2 protocol with 20 seeds, paired tests, multiplicity correction, effect sizes and confidence intervals, and with the architecture selected on validation data alone, the model reaches a Macro-F1 of 0.357 ± 0.019 that is statistically indistinguishable from four standard baselines spanning 29K–118K parameters.

The contribution is therefore not accuracy. It is that a physiologically constrained, inspectable basis costs nothing in accuracy relative to standard architectures, that the constraint measurably helps and is substantially retained through training, and that a rarity-anchored curriculum delivers a large and reproducible gain on the supraventricular class that the inter-patient protocol makes hardest. The same protocol also produced findings that a less rigorous one would have missed: that self-attention over a five-element interval vector is actively harmful, that the choice of mother wavelet within the wavelet-KAN family is not consequential, that the wavelet- and B-spline-basis KANs share a specific supraventricular-to-ventricular bias on supraventricular-enriched data, and that the model’s rhythm features depend significantly on information from beats that have not yet occurred. We report the negative and diagnostic results alongside the positive ones because, on this evidence, they are the more transferable half of what the study established.

ACKNOWLEDGMENT

The authors acknowledge the Vellore Institute of Technology (VIT) for funding the Article Processing Charge of this manuscript.

Ethics: This study uses only publicly available, fully de-identified ECG databases distributed by PhysioNet [23]. No new human-subject data were collected and no institutional review board approval was required.

Conflict of Interest: The authors declare no competing financial or non-financial interests.

CRedit Author Statement: Venkateramanan Manivannan: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. Ramanathan Lakshmanan: Conceptualization, Resources, Validation, Supervision, Project

Administration, Funding Acquisition, Writing – Review & Editing.

Data Availability: All datasets analysed are publicly available from PhysioNet [23]: the MIT-BIH Arrhythmia Database [22] (<https://physionet.org/content/mitdb/1.0.0/>), the St Petersburg INCART 12-lead Arrhythmia Database (<https://physionet.org/content/incartdb/1.0.0/>), the MIT-BIH Supraventricular Arrhythmia Database [25] (<https://physionet.org/content/svdb/1.0.0/>), and PTB-XL [26] (<https://physionet.org/content/ptb-xl/1.0.3/>).

Code Availability: Source code, preprocessing scripts, preseed result files and trained weights are available at <https://github.com/vrhrsr/WavKAN-CL>.

- [1] J. Takalo-Mattila, J. Kiljander, and J.-P. Soininen, "Inter-patient ECG classification using deep convolutional neural networks," *Proceedings of the 21st Euromicro Conference on Digital System Design (DSD)*, pp. 421–425, 2018.
- [2] L. Guo, G. Sim, and B. Matuszewski, "Inter-patient ECG classification with convolutional and recurrent neural networks," *Biocybernetics and Biomedical Engineering*, vol. 39, no. 3, pp. 868–879, 2019.
- [3] T. Akan, S. Alp, and M. A. N. Bhuiyan, "ECGformer: A transformer-based ensemble technique for cardiac arrhythmia classification," *arXiv preprint arXiv:2401.05434*, 2024.
- [4] R. Bahrami and A. M. Fotouhi, "Investigation of inter-patient, intra-patient, and patient-specific based training in deep learning for classification of heartbeat arrhythmia," *Cardiovascular Engineering and Technology*, vol. 16, pp. 333–342, 2025.
- [5] Q. Xiao, K. Lee, S. A. Mokhtar, I. Ismail, A. L. b. M. Pauzi, Q. Zhang, and P. Y. Lim, "Deep learning-based ECG arrhythmia classification: A systematic review," *Applied Sciences*, vol. 13, no. 8, p. 4964, 2023.
- [6] G. A. L. Silva *et al.*, "A systematic review of ECG arrhythmia classification: Adherence to standards, fair evaluation, and embedded feasibility," *arXiv preprint arXiv:2503.07276*, 2025.
- [7] P. De Chazal, M. O'Dwyer, and R. B. Reilly, "Automatic classification of heartbeats using ECG morphology and heartbeat interval features," *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 7, pp. 1196–1206, 2004.
- [8] A. Taleban, R. Sparapani, P. Noffke, S. Zlochiver, Q. Lu, M. E. Widlansky, and J. Luo, "Explainable artificial intelligence in electrocardiography: A systematic review," *Biomedical Signal Processing and Control*, vol. 114, p. 109325, 2025.
- [9] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark, "KAN: Kolmogorov–Arnold networks," *arXiv preprint arXiv:2404.19756*, 2024.
- [10] Z. Bozorgasl and H. Chen, "Wav-KAN: Wavelet Kolmogorov–Arnold networks," *arXiv preprint arXiv:2405.12832*, 2024.
- [11] P. S. Addison, "Wavelet transforms and the ECG: a review," *Physiological Measurement*, vol. 26, no. 5, pp. R155–R199, 2005.
- [12] C. Zhao, B. Lai, Y. Xu, Y. Wang, and H. Dong, "MAK-Net: A multi-scale attentive Kolmogorov–Arnold network with BiGRU for imbalanced ECG arrhythmia classification," *Sensors*, vol. 25, no. 13, p. 3928, 2025.
- [13] F.-y. Zhou, Y.-h. Sun, and Y.-w. Wang, "Inter-patient ECG arrhythmia heartbeat classification network based on multiscale convolution and FCBA," *Biomedical Signal Processing and Control*, vol. 89, p. 105789, 2024.
- [14] S. Mousavi and F. Afghah, "Inter- and intra-patient ECG heartbeat classification for arrhythmia detection: A sequence to sequence deep learning approach," in *2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2019, pp. 1308–1312.
- [15] S. Rigas, D. Verma, G. Alexandridis, and Y. Wang, "Initialization schemes for Kolmogorov–Arnold networks: An empirical study," *arXiv preprint arXiv:2509.03417*, 2025.
- [16] Y. Bengio, J. Louradour, R. Collobert, and J. Weston, "Curriculum learning," in *Proceedings of the 26th International Conference on Machine Learning (ICML)*. Montreal, Canada: ACM, 2009, pp. 41–48.
- [17] J. Bae and J. Shin, "Handling imbalanced medical dataset with continuous class features using improved contrastive learning," in *Proceedings of the IEEE International Conference on Consumer Electronics (ICCE)*. IEEE, 2025.
- [18] S. Schmale, P. Hempel, and N. Spicher, "Curriculum learning using real and simulated data in deep learning models for electrocardiography classification," in *Proceedings of the 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 2025.
- [19] J. Dai, Q. Zhang, C. Dai, D. Lan, and C. Han, "Angular gaussian supervised contrastive learning for long-tailed electrocardiogram arrhythmia diagnosis," *arXiv preprint arXiv:2607.14613*, 2026.
- [20] M. M. Farag, "A tiny matched-filter-based CNN for inter-patient ECG classification and arrhythmia detection at the edge," *Sensors*, vol. 23, no. 3, p. 1365, 2023.
- [21] D. H. Elsheikh, A. S. Hassan, N. M. Yhiea, A. M. Fareed, and E. A. Rashed, "Lightweight deep learning architecture for multi-lead ECG arrhythmia detection," *Sensors*, vol. 25, no. 17, p. 5542, 2025.
- [22] G. B. Moody and R. G. Mark, "The impact of the MIT-BIH Arrhythmia Database," *IEEE Engineering in Medicine and Biology Magazine*, vol. 20, no. 3, pp. 45–50, 2001.
- [23] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.
- [24] E. J. d. S. Luz, W. R. Schwartz, G. Cámara-Chávez, and D. Menotti, "ECG-based heartbeat classification for arrhythmia detection: A survey," *Computer Methods and Programs in Biomedicine*, vol. 127, pp. 144–164, 2016.
- [25] S. D. Greenwald, "Improved detection and classification of arrhythmias in noise-corrupted electrocardiograms using contextual information," Ph.D. dissertation, Harvard–MIT Division of Health Sciences and Technology, 1990.
- [26] P. Wagner, N. Strodthoff, R.-D. Boussejot, D. Kreiseler, F. I. Lunze, W. Samek, and T. Schaeffter, "PTB-XL, a large publicly available electrocardiography dataset," *Scientific Data*, vol. 7, no. 1, p. 154, 2020.
- [27] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.
- [28] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [29] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Hillsdale, NJ: Lawrence Erlbaum Associates, 1988.

Venkateramanan Manivannan received the B.E. degree in Computer Science and Engineering and is currently pursuing the M.Tech degree at the School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology (VIT), Vellore, India.

His research interests include deep learning, neural network architectures, explainable AI, biomedical signal processing, and computationally efficient models for edge deployment, with a current focus on interpretable and resource-efficient frameworks for healthcare applications.

Ramanathan Lakshmanan received the B.E. degree from Bharathidasan University, Tiruchirappalli, India, the M.E. degree from Sathyabama University, Chennai, India, and the Ph.D. degree from VIT University, Vellore, India, all in Computer Science and Engineering. He is currently a Professor with the Department of IoT, School of Computer Science and Engineering, VIT, Vellore, India, with over 18 years of teaching and research experience.

His research interests include data science, cyber-physical systems, fog and edge computing, IoT, and machine learning. He serves as an Associate Editor for Network: Computation in Neural Systems.